RNA-seq Best Practices & The Pipeline

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DSCI 512: RNA sequencing data analysis

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1. Best practices In experimental design

Experimental design in RNA-seq projects is very import and should be driven by the biological question

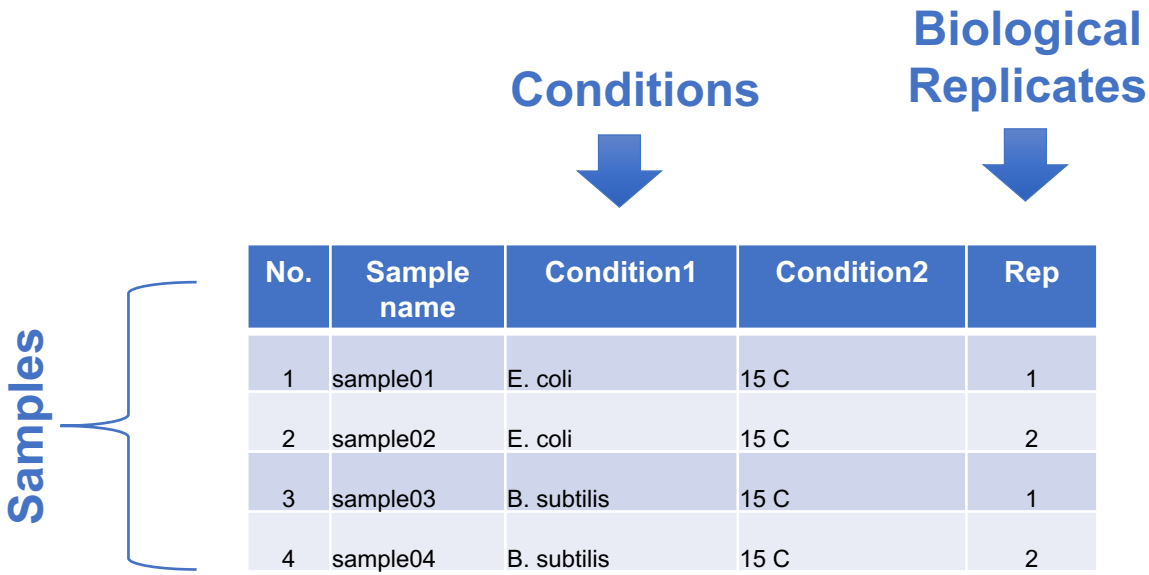
- Biological question?
  - Conditions?
  - What characteristics of the transcriptome do you expect will be different?
  - What will you do next with this information?

- Resources?
  - How many ***biological*** replicates?
  - How much material?
  - How much funding?

- How can batch effects be reduced?

- Who will analyze the data?

On vocabulary – samples, conditions, replicates



No.	Sample name	Condition1	Condition2	Rep
1	sample01	E. coli	15 C	1
2	sample02	E. coli	15 C	2
3	sample03	B. subtilis	15 C	1
4	sample04	B. subtilis	15 C	2

- **Samples** – Each RNA-seq library that was prepped and sequenced is a sample
- **Conditions** – Represent the differences we are trying to assess
  - Can be pairwise such as mutant v. wild-type, disease v. healthy
  - Can be ordered – developmental time, temperature, salt conc.
- **Replicates** – these are duplicates
  - Biological replicates – different organisms were harvested. Very important.
  - Technical replicates – the same organisms were harvested. Not advised.

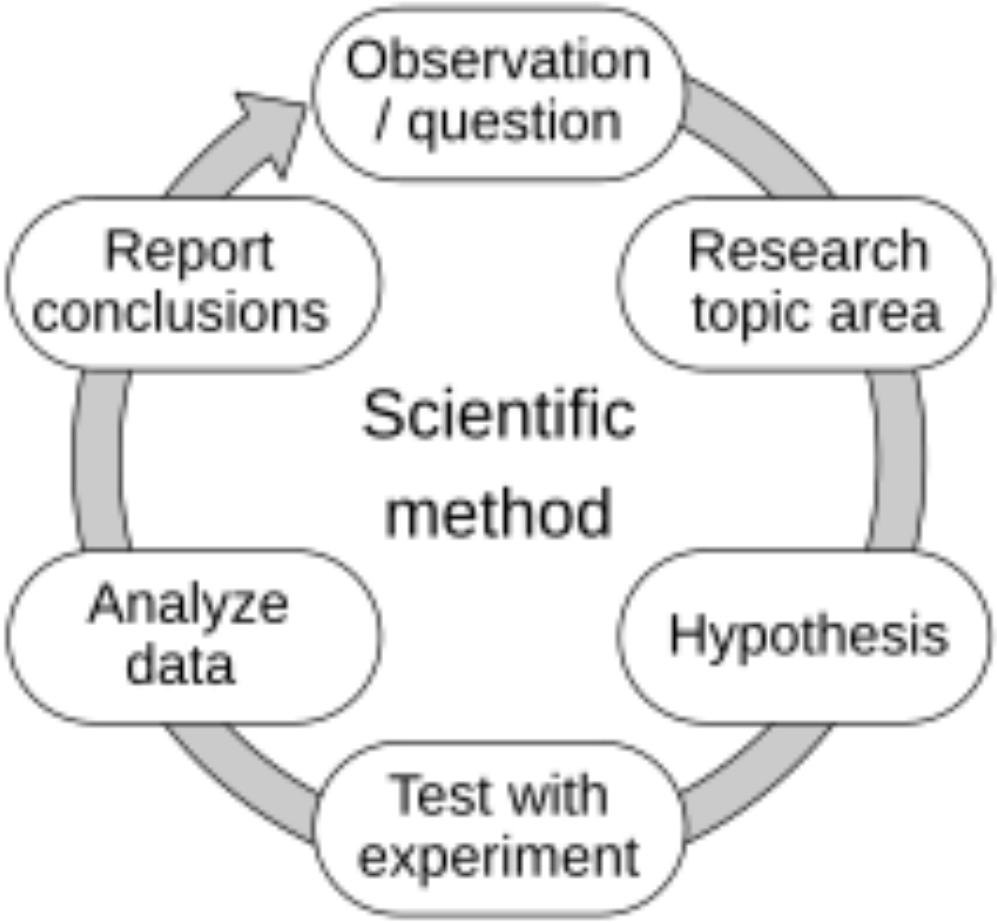
Ideal practices

- A minimum of ***FOUR*** biological replicates per condition for differential abundance experiments
- ***Defend against batch effects*** when performing biological replicates
- Consult with your sequencing facility and analytical team ***BEFORE*** doing the experiment
- Think deeply about positive and negative ***controls***

Key limitations and assumptions of RNA-seq

- For differential expression analysis, protocols ***assume*** at least ***90 % of RNAs are NOT changing*** in abundance between conditions
- mRNA abundance is ***not always a good marker for protein*** production or transcriptional activity
- Most samples are mixture of ***heterogeneous*** cells or tissue types
- Differences between conditions can be ***direct or indirect*** effects
- Individual cells are inherently ***variable***
- Some important genes are ***lowly expressed*** or have ***modest changes*** in RNA-seq

Where does transcriptome profiling fit in the scientific method?



Have a plan for the next steps

- **Confirm** results with an alternative method
  - Microscopy
  - GFP reporter
  - Western blot
- **Zoom in** on key genes to study
  - Build a story
- **Follow up** on the hypotheses generated